



Faculty Of Computing & Software Engineering

G4 -Computer Science

➤ Introduction to Data Mining and Data Warehousing

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Lecture-1

➤ Introduction to Data Mining

Objectives

At the end of the chapter the student be able to understand:-

- ❖ Concepts of Data Mining, Data Warehousing
- ❖ Challenges in Data Mining
- ❖ Application of data mining
- ❖ Data mining functionality
- ❖ Are all the patterns interesting?

What is data mining?

- Data mining is an interdisciplinary subfield of computer science and statistics with an overall goal of extracting information (with intelligent methods) from a data set and transforming the information into a comprehensible structure for further use. **(Wikipedia)**
- Data Mining refers to the process of analyzing large databases to find the interesting (non-trivial, implicit, previously unknown and potentially useful) information or pattern.

- **Alternative names:**
 - ✓ *Knowledge discovery(mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, business intelligence, etc.*

4/28/2026

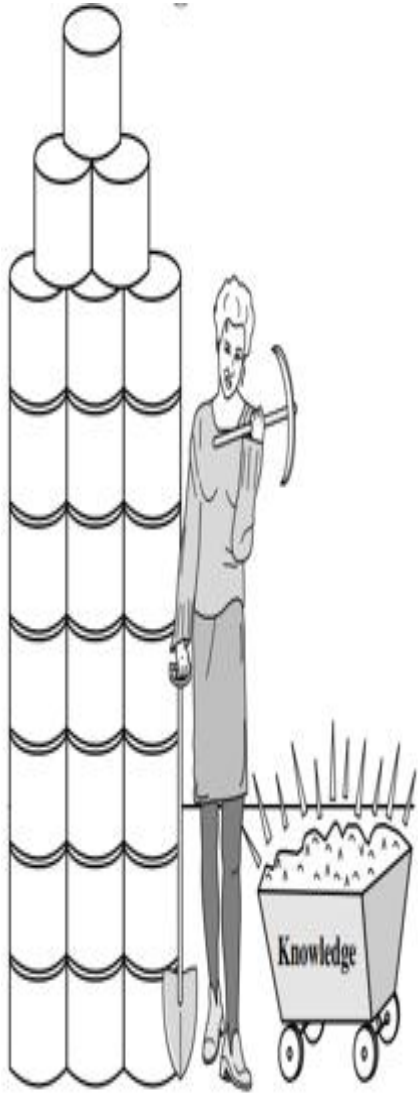


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❖ Data mining is the process of analyzing large databases using various techniques to find patterns in data that are:

- ✓ **valid**: not only represent current state, but also hold on new/test data with some certainty
- ✓ **novel**: non-obvious to the system that are generated as new facts
- ✓ **useful**: should be possible to act on the item or problem
- ✓ **understandable**: humans should be able to interpret the pattern

How Data Mining works?



- ✓ Data mining is looking for hidden, valid, and potentially useful patterns in huge data sets. Data Mining is all about discovering unsuspected/ previously unknown relationships amongst the data.
- ✓ The process of **uncovering trends, common themes or patterns in “big data”**.
- ✓ It is the ideas from statistics and computer science (**machine learning and database methods**) applied to large databases in science, engineering and business.

Query Examples

✓ Database

- ✓ Find all credit applicants with last name of Smith.
- ✓ Identify customers who have purchased more than \$10,000 in the last month.
- ✓ Find all customers who have purchased milk

✓ Data Mining

- Find all credit applicants who are poor credit risks.
(classification)
- Identify customers with similar buying habits. (Clustering)
- Find all items which are frequently purchased with milk.
(frequent pattern analysis)

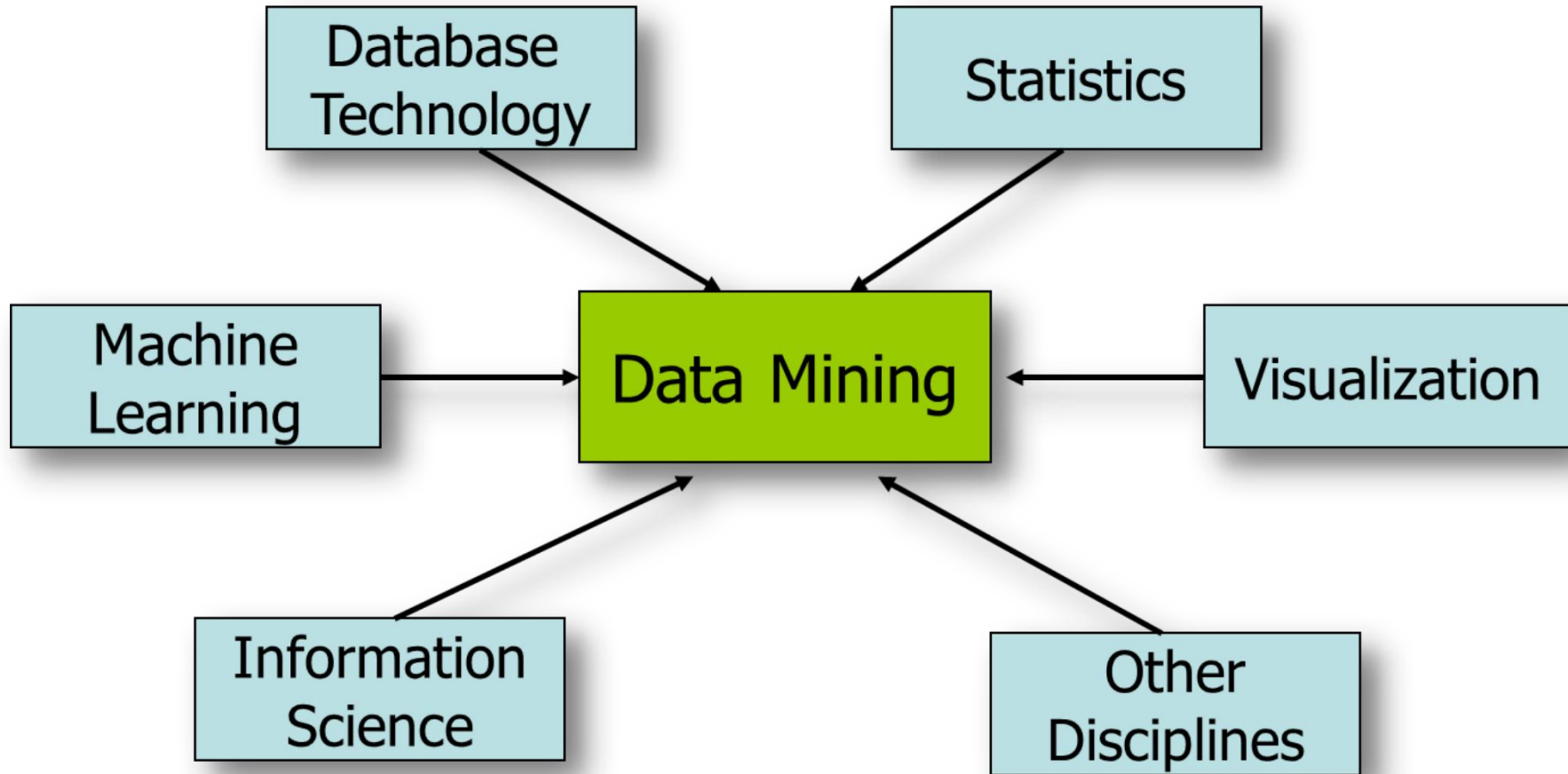
Why Data Mining?

- ✗ Wide availability of huge amounts of data
 - ✗ Need for turning such data into useful information and knowledge
 - ✗ Large supplies of powerful and affordable computer
 - ✗ A huge number of databases and information repositories available for transaction management, information retrieval and data analysis.
- ✗ In addition, consider expert system technologies, which typically rely on users or domain experts to manually input knowledge into knowledge bases. Unfortunately, this procedure is prone to biases and errors, and is extremely time-consuming and costly. Data mining tools perform data analysis and may uncover important data patterns, contributing greatly to business strategies, knowledge bases, and scientific and medical research.
- ✗ The information and knowledge gained can be used for applications ranging from market analysis, fraud detection, and customer retention, to production control and science exploration.

❖ Major sources of abundant data

- ✓ **Business:** e-commerce, transactions, stocks, ...
- ✓ **Science:** Remote sensing, bioinformatics, scientific simulation, ...
- ✓ **Society and everyone:-** news, digital cameras, YouTube, social media, mobile devices, ...
- ✓ **Applications:** Banking, Distributed systems etc.
- ✓ **WWW:** refers to a collection of documents and resources like audio, video, text, etc which are identified by Uniform Resource Locators (URLs) through web browsers, linked by HTML pages, and accessible via the Internet network.
- ✓ etc

Data Mining: Confluence of Multiple Disciplines



Data Mining Tasks and Applications

- ❖ What techniques people apply on data?
- ❖ They apply data mining techniques, algorithms and discover useful knowledge.
- Data mining tasks are broadly classified into two categories:

1. Descriptive Tasks:-

Goal: Find patterns in the data.

- ❖ Example: *Which products are often bought together?*

2. Predictive Tasks:-

Goal: Predict unknown values of a variable given observations (e.g., from the past)

- ❖ Example: *Will a person click a online advertisement?* given her browsing history

Data mining technique/Tasks

Predictive

- Classification
- Prediction
- Time series analysis

Descriptive

- Cluster analysis
- Association rules

Classification

Goal: Previously unseen records should be assigned a class from a given set of classes as accurately as possible.

- Construct a models based on some training examples (pre-labeled (classified) instances).
- Predict some unknown class labels.

Approach:-

- ✓ Given a collection of records (*training set*)
 - ❖ each record contains a set of *attributes*
 - ❖ one attribute is the *class attribute (label)* that should be predicted
- ✓ Find a *model (construct model)* for predicting the class attribute as a function of the values of other attributes.



Classification: Example

- Training set:



"tree"



"tree"



"tree"



"not a tree"



"not a tree"



"not a tree"

- Learned model: "Trees are big, green plants without wheels."

Classification

❖ Direct Marketing

❖ **Goal**: Reduce cost of mailing by targeting a set of consumers likely to buy a new cell-phone product.

❖ **Approach**:

- ❖ Use the data for a similar product introduced before.
- ❖ We know which customers decided to buy and which decided otherwise.
- ❖ This {buy, don't buy} decision forms the class attribute.
 - ✓ Collect various demographic, lifestyle, and company-interaction related information about all such customers.
 - ✓ age, profession, location, income, marriage status, visits, logins, etc.
 - ✓ Type of business, where they stay, how much they earn, etc.
- ❖ Use this information as input attributes to learn a classification model
- ❖ Apply model to decide which consumers to target

Classification.....

- **Application area:** Fraud Detection
- **Goal:** Predict fraudulent cases in credit card transactions.
- **Approach:**
 - ✓ Use credit card transactions and information about account-holders as attributes
 - ❖ When and where does a customer buy? What does he buy?
 - ❖ How often he pays on time? etc.
 - ✓ Label past transactions as fraud or fair transactions This forms the class attribute
 - ✓ Learn a model for the class attribute from the transactions
 - ❖ Use this model to detect fraud by observing credit card transactions on an account



Time Series Analysis

- ✓ With Time Series analysis, the value of an attribute is examined as it varies over time.
- ✓ This concerns the analysis of data collected over time ... weekly values, monthly values, quarterly values, yearly values, etc.
- ❖ Examples of time series include the continuous monitoring of a person's heart rate, hourly readings of air temperature, daily closing price of a company stock, monthly rainfall data, and yearly sales figures.



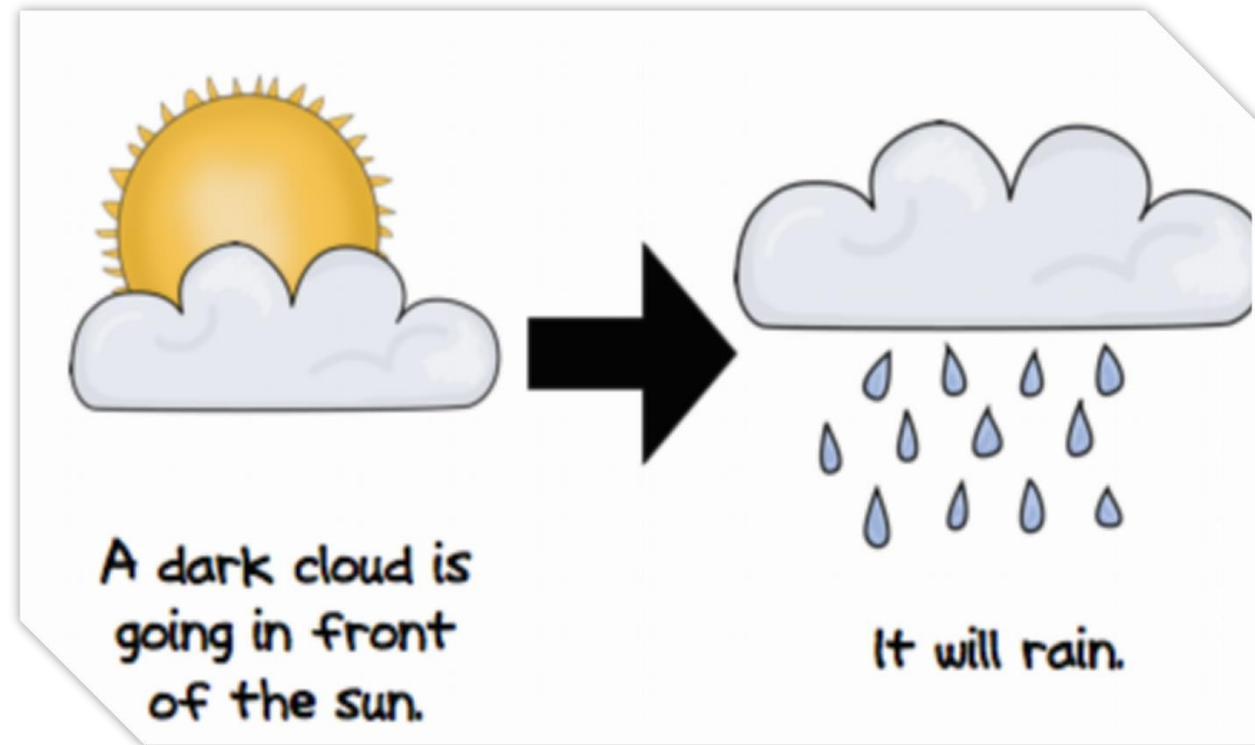
Daily Stock Price



Weekly interest rates

Prediction

- ❖ Many real world data mining applications can be seen as predicting future states based on past and current data.



Clustering

- Clustering is similar to classification except that the groups are not pre-defined, but rather defined by the data alone.
- Clustering is alternatively referred to as unsupervised learning or segmentation.



Association Rules

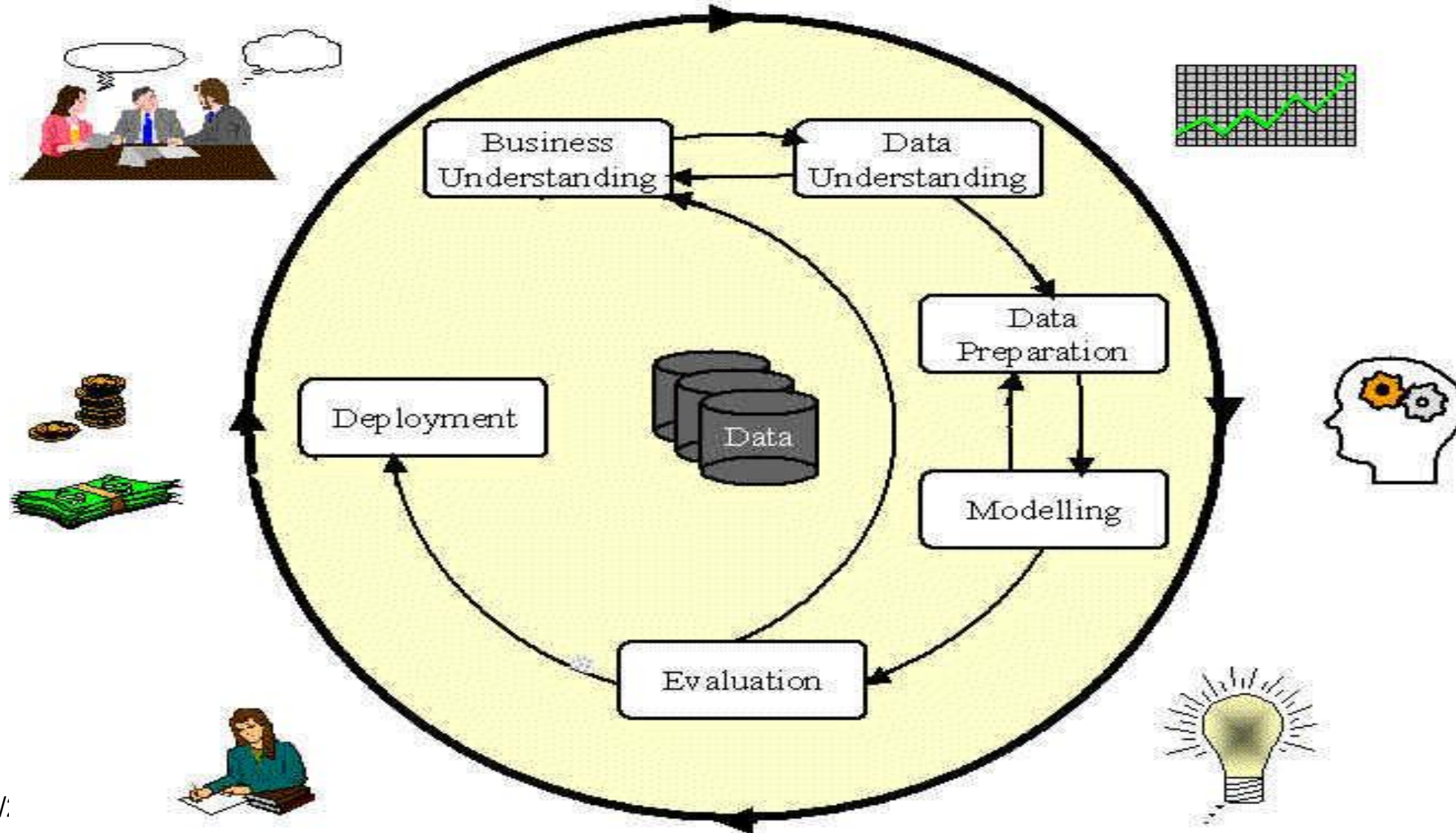
- ✓ Association rule is a model that identifies specific types of data associations.
- ✓ Association rules are **if-then** statements that help to show the probability of relationships between data items within large data sets.



Stages in DM: The KDD process

- ✓ **Selection:** Obtain data from various heterogeneous sources such as databases, data warehouses, files, non-electronic records, etc.
- ✓ **Preprocessing:** Cleanse inconsistent & incorrect data; fills incomplete records; predict missing values; correct erroneous & anomalous data.
- ✓ **Transformation:** Convert data from different sources into common new format. Apply data reduction & data categorization to ease data mining
- ✓ **Mining:** apply classification/clustering techniques to obtain predictive / descriptive models.
- ✓ **Interpretation/Evaluation:** Present results to user in meaningful manner using various visualization and GUI strategies.

DM Model: Cross Industry Standard Process for Data Mining (CRISP-DM)



Are All of the Patterns Interesting?

- ❖ A data mining system has the potential to generate thousands or even millions of patterns, or rules.
 - “are all of the patterns interesting?”
 - Typically not only a small fraction of the patterns potentially generated would actually be of interest to any given user.
- ❖ A pattern is interesting if it is:
 - ✓ easily understood by humans,
 - ✓ valid on new or test data with some degree of certainty,
 - ✓ potentially useful,
 - ✓ novel.
- ❖ An interesting pattern represents knowledge.

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- Several objective measures of pattern interestingness exist, example:
 - ✓ An objective measure for **association rules** of the form $X \Rightarrow Y$ is rule **support**,
 - ✓ Another objective measure for association rules is **confidence**,
 - ✓ In general, each interestingness measure is associated with a threshold, which may be controlled by the user.
- ❖ Accuracy in classification
 - ✓ Analyze true positive and false positive to calculate recall, precision of the system
 - ✓ Measure percentage of correct classification

Data Mining implementation issues

- ❖ **Scalability:** As the volume and complexity of data continue to grow exponentially, data mining algorithms and techniques must be scalable to handle large datasets. Developing efficient algorithms that can process massive amounts of data in a reasonable time frame remains a significant challenge.
- ❖ **High-dimensional data:** Many real-world datasets have a high number of dimensions, which can lead to the curse of dimensionality. As the number of dimensions increases, the sparsity of data increases, making it difficult to discover patterns and extract meaningful information. Developing effective methods for dimensionality reduction and feature selection is an ongoing challenge.
- ❖ **Privacy and ethical concerns:** Data mining often involves working with sensitive and personal information, raising concerns about privacy and ethical considerations.

Data Mining implementation issues

- ❖ **Noise and uncertainty:** Real-world datasets are often noisy, incomplete, and contain missing values or outliers. Dealing with these imperfections and uncertainties is a significant challenge in data mining research. Developing robust algorithms that can handle noisy data and provide reliable results is crucial.
- ❖ **Interpretable and explainable models:** While achieving high prediction accuracy is important, understanding and interpreting the underlying patterns and relationships in the data is equally crucial, especially in domains like healthcare and finance. Developing interpretable and explainable models that can provide insights into the decision-making process is an ongoing challenge.
- ❖ **Handling complex data types:** Data mining techniques need to handle diverse and complex data types, such as text, images, graphs, and time-series data. Developing algorithms that can effectively mine information from these data types and integrate them into meaningful insights remains a challenge.
- ❖ **Real-time and streaming data:** With the advent of the Internet of Things (IoT) and streaming data sources, data mining techniques need to adapt to handle real-time data streams. Processing data in real-time poses unique challenges, such as time constraints, data arrival order, and concept drift detection.

Data Mining implementation issues

- ❖ **Incorporating domain knowledge:** Domain expertise and prior knowledge often play a crucial role in data mining tasks. However, effectively incorporating domain knowledge into the mining process and leveraging it to improve accuracy and interpretability is challenging. Developing methods that can seamlessly integrate domain knowledge with data mining algorithms is an active area of research.
- ❖ **Bias and fairness:** Data mining models can unintentionally maintain existing biases present in the data, leading to unfair outcomes and discrimination.
- ❖ **Reproducibility and evaluation:** Ensuring the reproducibility of results and evaluating the performance of data mining algorithms on different datasets are essential for advancing the field. However, achieving reproducibility and establishing standardized evaluation metrics and benchmarks remain challenging tasks.
- **These challenges highlight the ongoing research efforts in data mining to develop more robust, scalable, interpretable, and fair techniques that can handle complex, real-world data effectively.**

Application of data mining

❖ Business and Market Analysis:

- to identify patterns and trends in customer behavior,
- market segmentation,
- product recommendations,
- sales forecasting, and identifying cross-selling and upselling opportunities.

❖ Healthcare and Medicine:

- analyze medical records, patient data, and clinical trials to identify patterns and correlations that can help
 - ✓ in disease diagnosis,
 - ✓ treatment prediction,
 - ✓ patient monitoring,
 - ✓ drug discovery, and healthcare decision-making.

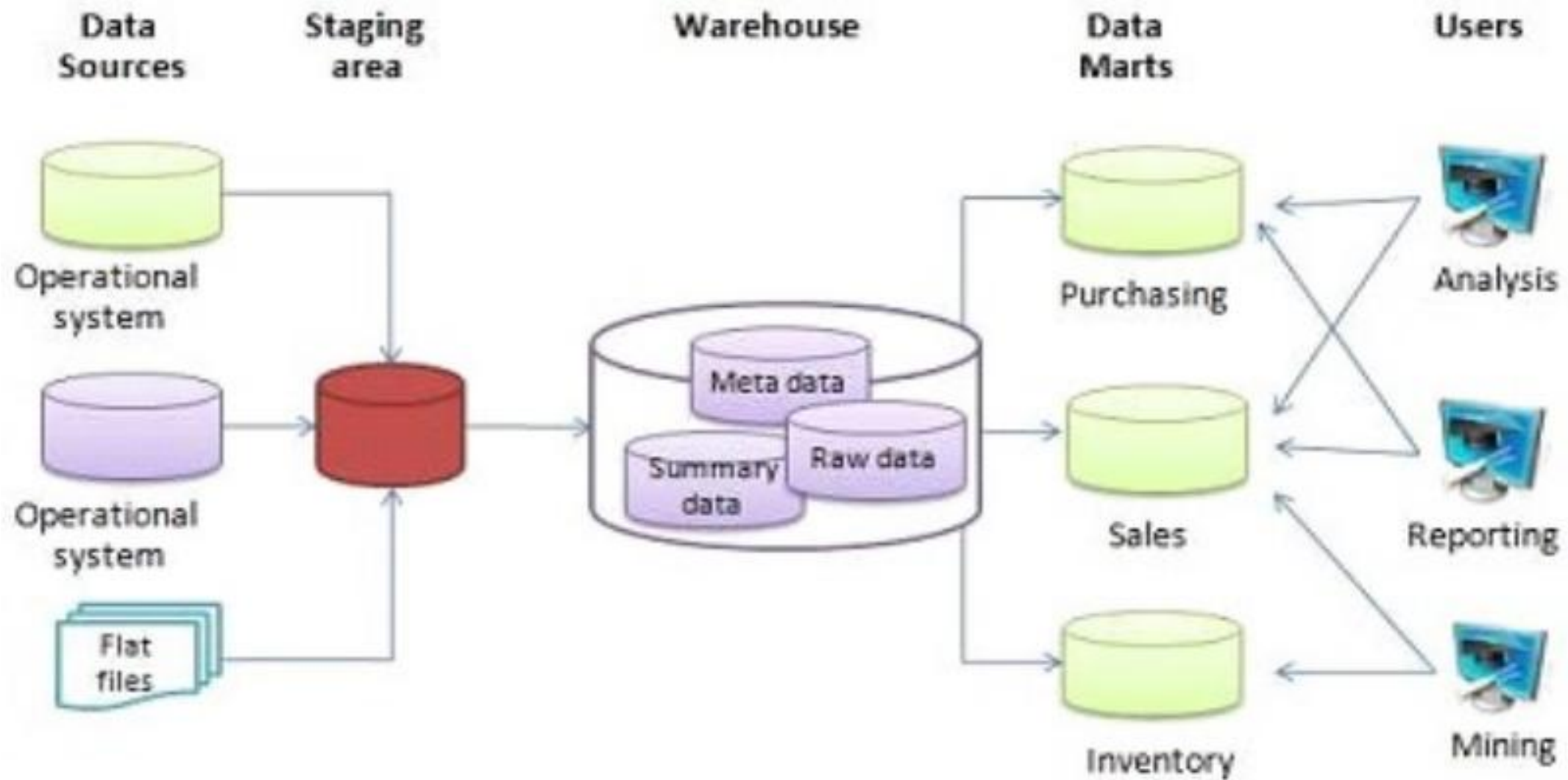
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- ❖ **Fraud Detection and Risk Management:** Data mining techniques are employed in financial institutions and insurance companies
 - to detect fraudulent activities,
 - ❖ such as credit card fraud,
 - ❖ insurance fraud, and money laundering.
 - ❖ It also helps in risk assessment and management by identifying potential risks and predicting future trends.
- ❖ **Education:**
 - ❖ used to analyze student performance,
 - ❖ identify at-risk students, personalize learning paths,
 - ❖ and improve educational outcomes. It helps in adaptive learning, course recommendation, and early intervention.

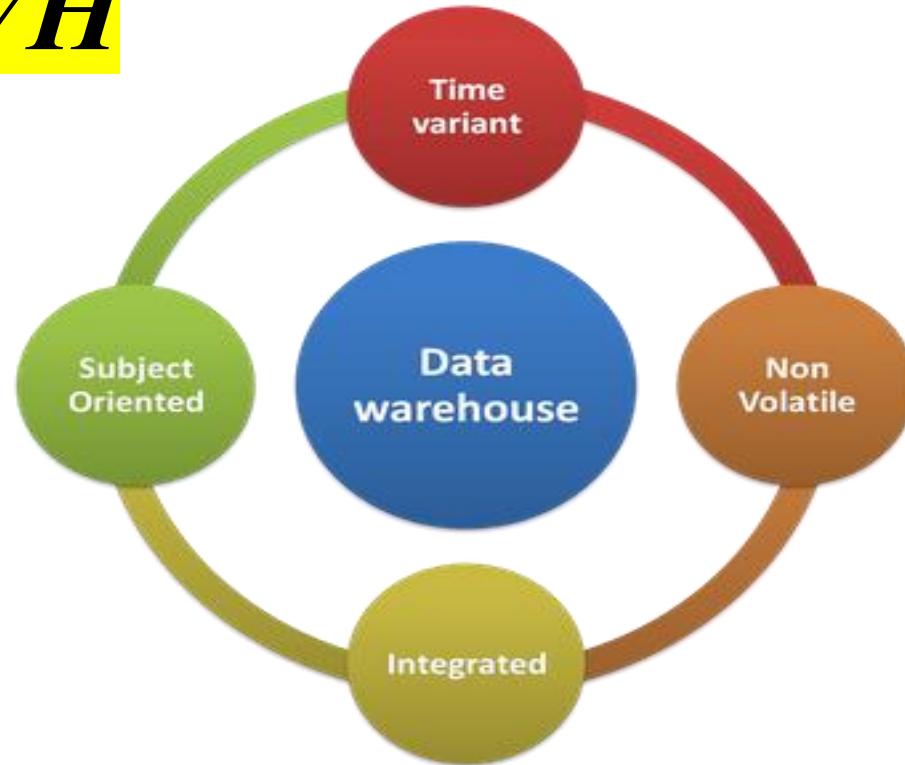
Lecture 2: Data warehousing

Overview of data warehousing

- ❖ **Data Warehousing** is the process of collecting, storing, and managing large amounts of data from different sources so it can be easily analyzed.
- ❖ A **data warehouse** is a **repository** of information collected from multiple sources, stored under a unified schema.
- ❖ Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.



Characteristics of DWH

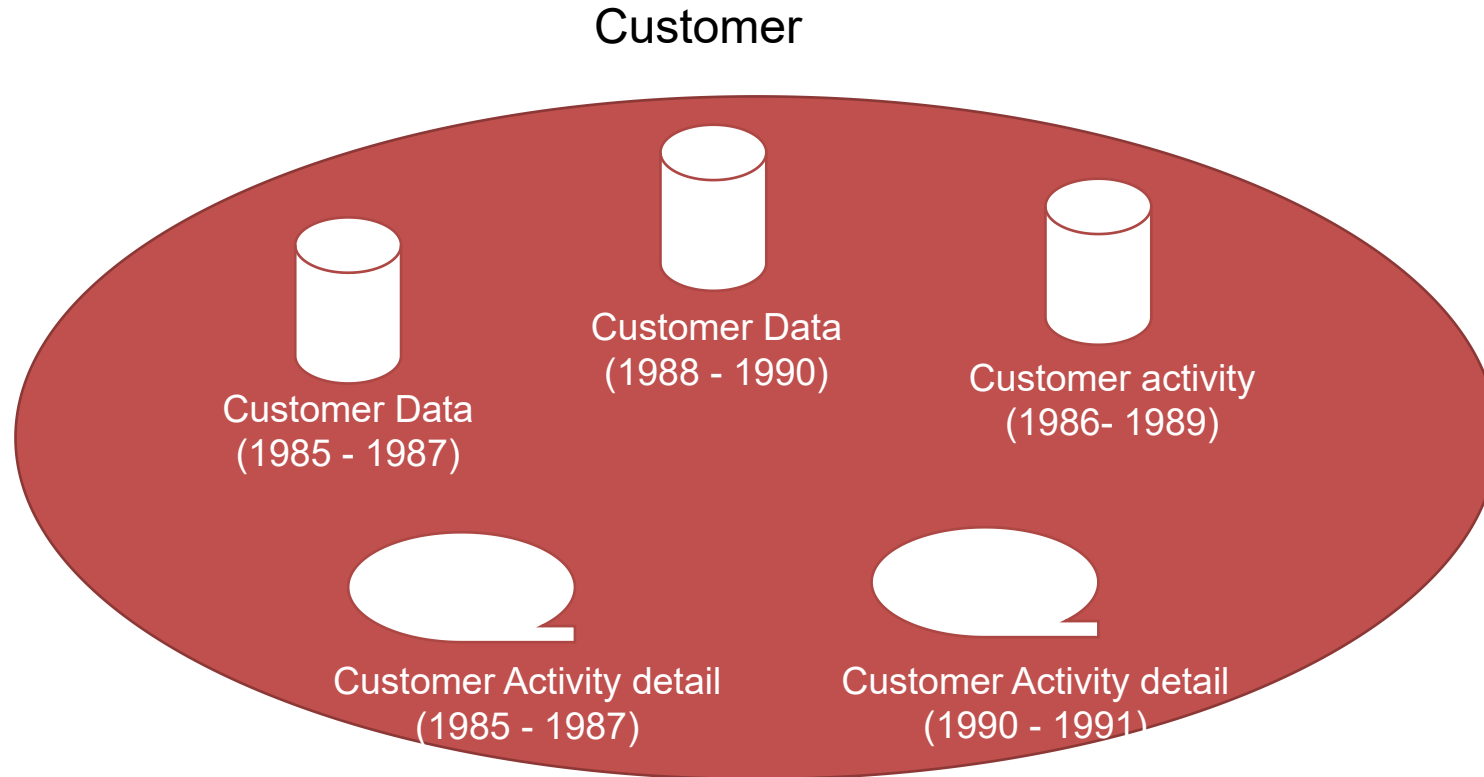


- ✓ A data warehouse is a *subject-oriented, Integrated, time-variant, non-volatile* collection of data in support of management's decision making process.
- ✓ It is generally used for **research and decision support**.

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❖ Subject Oriented:

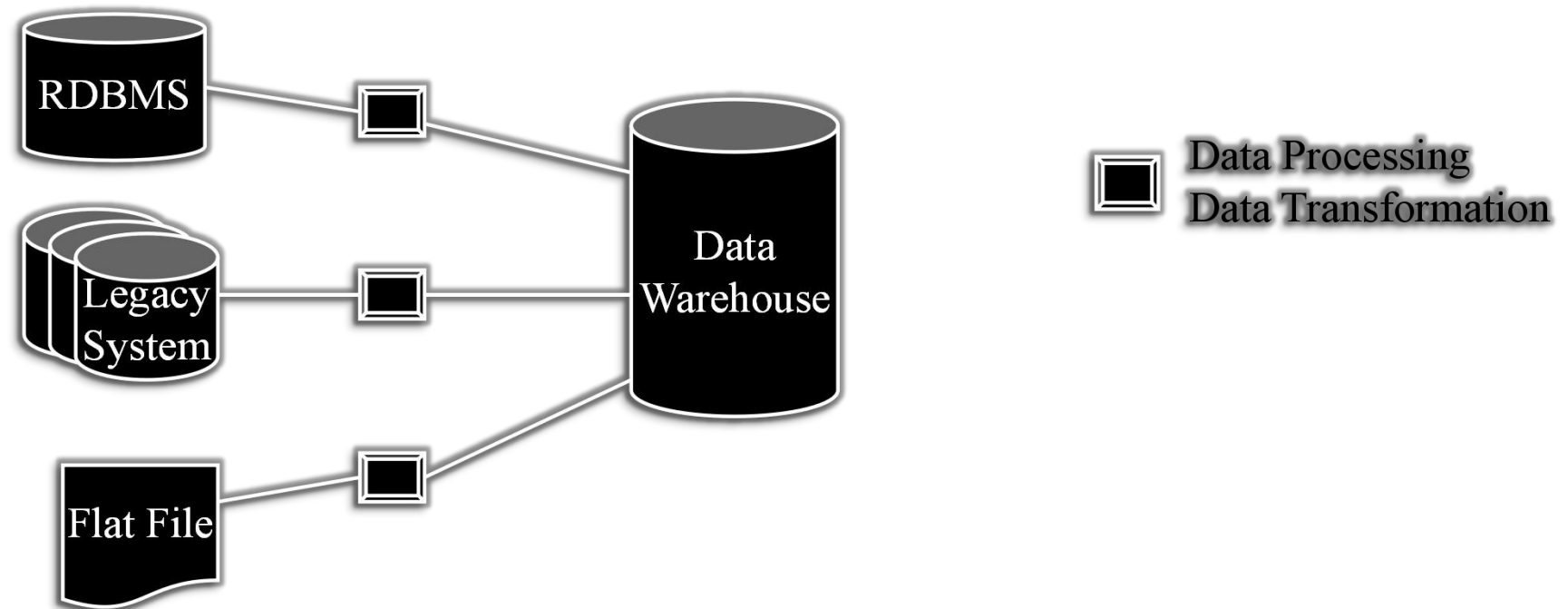
- ✓ Data warehouse is organized around subjects such as *sales*, *product*, *customer*



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❖ Integrated:

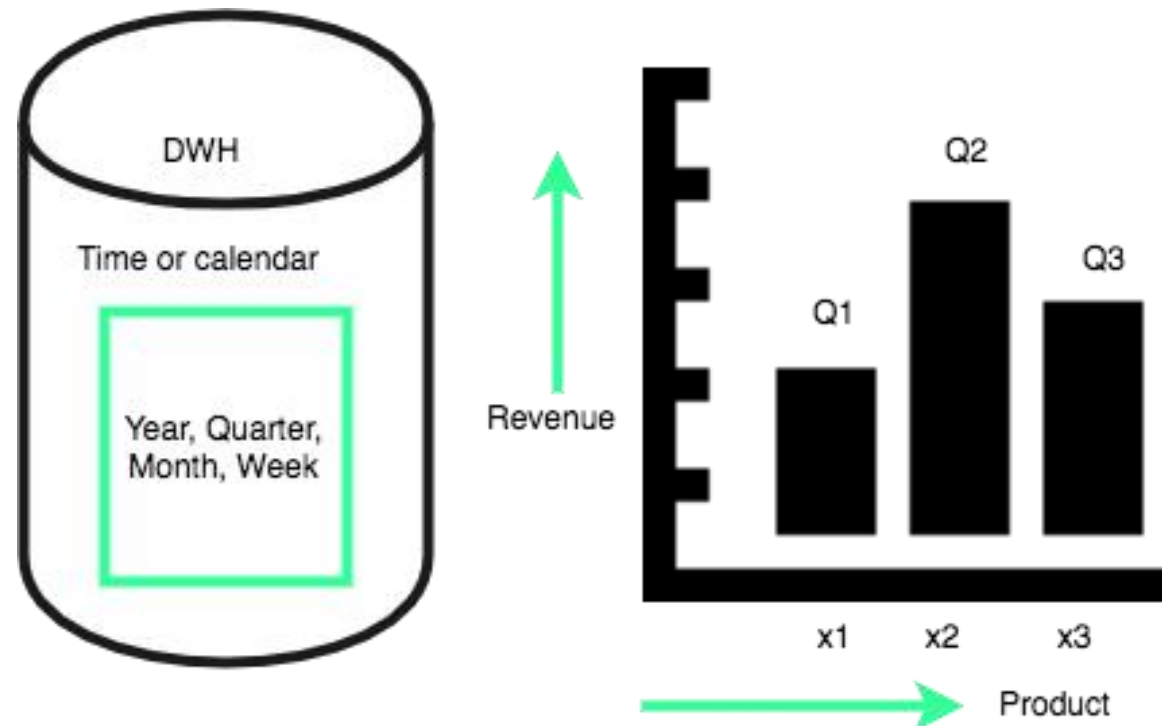
- ✓ Data Warehouse is constructed by integrating multiple heterogeneous sources.
- ✓ Data Preprocessing are applied to ensure consistency



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❖ Time Variant:

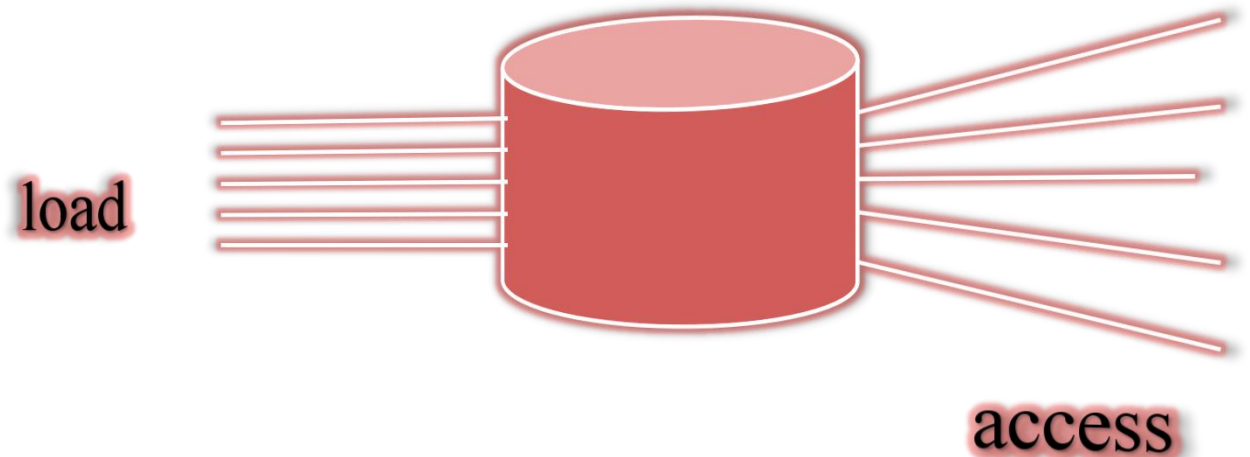
- ❖ Provides information from historical perspective e.g. past 5-10 years
- ❖ Data warehouse stores historical data.



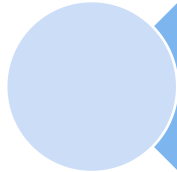
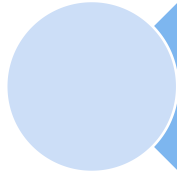
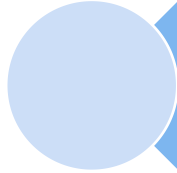
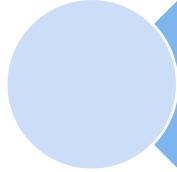
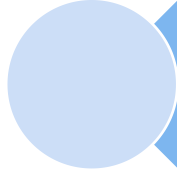
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❖ Non- volatile:

- ✓ Data once recorded cannot be updated.
- ✓ Data warehouse requires two operations:
 - ❖ Initial loading of data
 - ❖ Access of data



Need/importance of data warehouse

-  *source of information for report generation*
-  *Increase quality and flexibility of enterprise analysis*
-  *Ability to maintain better customer relationships*
-  *source for data analysis and data mining*
-  *More cost – effective decision making and policy formulation*

Features of Data warehouse

- ✓ It is separate from Operational Database.
- ✓ Integrates data from heterogeneous systems.
- ✓ Stores HUGE amount of data, more historical than current data.
- ✓ Queries are generally complex.
- ✓ Goal is to execute statistical queries and provide results which can influence decision making in favor of the Enterprise.

Differences between Operational Database Systems and Data Warehouses

- Because most people are familiar with commercial relational database systems, it is easy to understand what a data warehouse is by comparing these two kinds of systems.
- The major task of **on-line operational database systems** is to perform on-line transaction and query processing.
- These systems are called **on-line transaction processing (OLTP)** systems.
- They cover most of the day-to-day operations of an organization,
- such as purchasing, banking, payroll, registration, accounting etc.
- **Data warehouse systems**, on the other hand, serve users or knowledge workers in the role of data analysis and decision making.
- Such systems can organize and present data in various formats in order to accommodate the diverse needs of the different users.
- These systems are known as **on-line analytical processing (OLAP) systems**.

But, Why Have a Separate Data Warehouse

- ❖ Data warehouse queries are often complex and they present a general form of data.
- ❖ In contrast, OLTP systems require high concurrency, reliability, locking which provide good performance for short and simple queries.
- ❖ An operational database maintains current data. On the other hand, a data warehouse maintains historical data.
- ❖ Rather than using a query-driven approach, data warehousing employs an update-driven approach in which information from multiple, heterogeneous sources is integrated in advance and stored in a warehouse for direct querying and analysis.

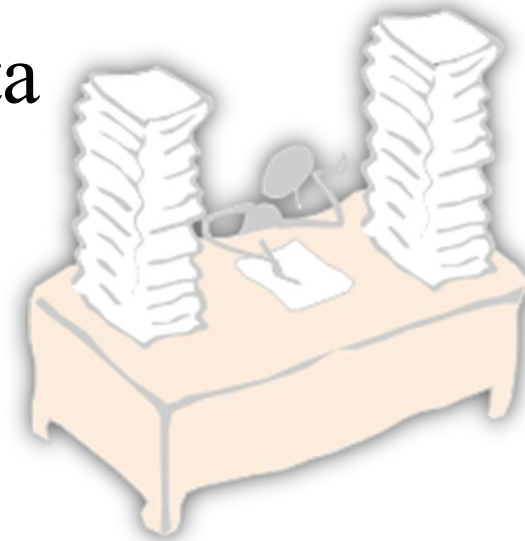
Data Warehouse for Decision Support & OLAP

- Putting Information technology to help the knowledge worker make faster and better decisions
 - ✓ What product promotions have the biggest impact on revenue?
 - ✓ How did the share price of software companies correlate with profits over last 10 years?
 - ✓ What products are sold? what product is not? when does sells go up?
 - ✓ Which product is not doing well in the market?

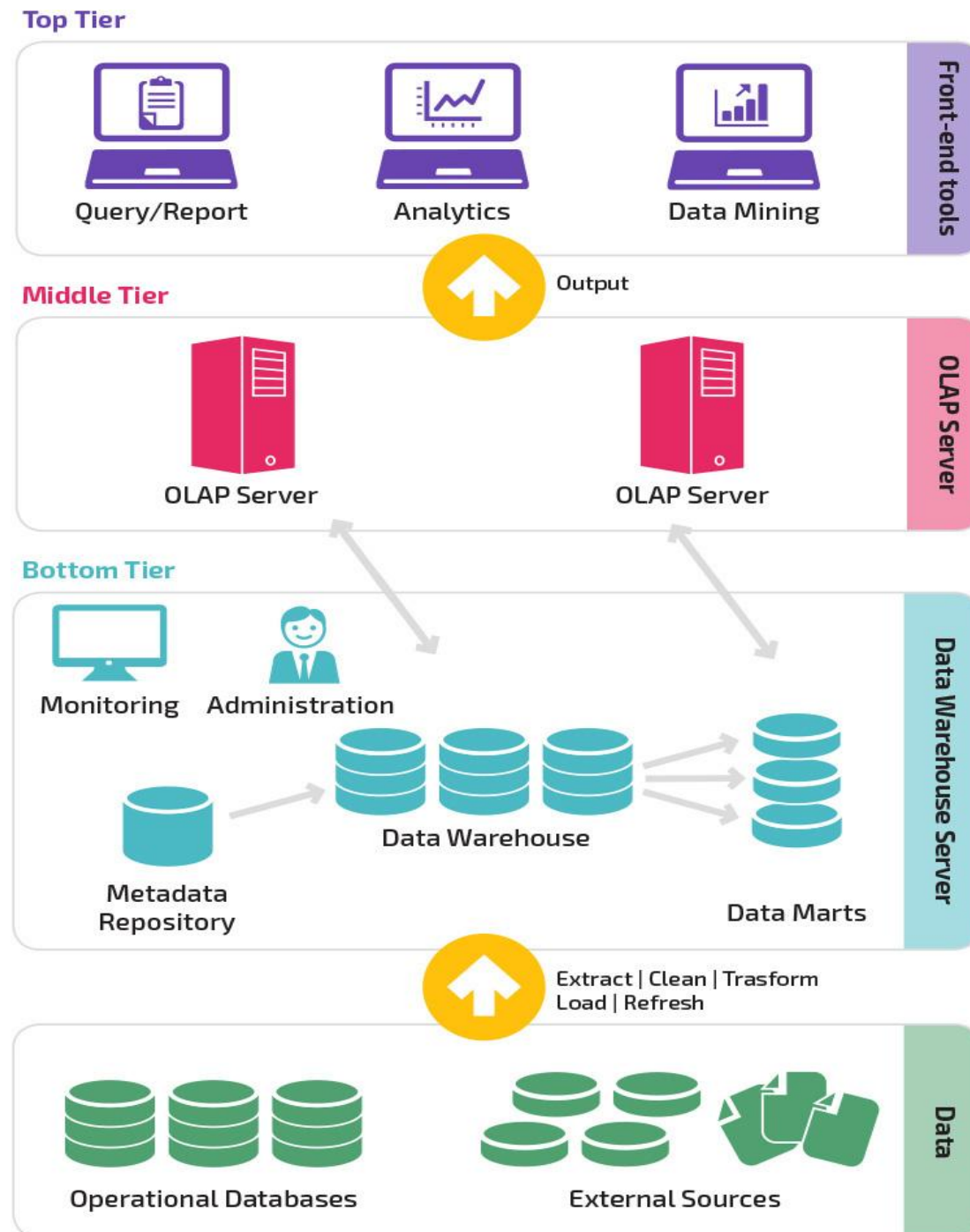
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❖ Data Mining works with Warehouse Data

- ✓ Data Warehousing provides the Enterprise with a memory
- ✓ Data Mining provides the Enterprise with intelligence



3 –Tier Data warehouse architecture



Trend analysis, prediction, classification and so on

- ✓ Implemented either ROLAP or MOLAP
- ✓ support and operate on multi-dimensional data structures

Almost always a relational DBMS, rarely flat files

Data Warehouse Models

❖ The enterprise warehouse

- An enterprise warehouse collects all of the information about subjects spanning the entire organization.
- It provides corporate-wide data integration,
- It typically contains detailed data as well as summarized data

❖ The data mart

- ✓ A data mart contains a subset of corporate-wide data
- ✓ The scope is confined to specific selected subjects.
- ✓ It can be either *Dependent* or *Independent* data mart.
- ✓ A subset of a data warehouse for small and medium-size businesses or departments within larger companies

➤ The virtual warehouse.

- ❖ is a set of views over operational databases.
- ❖ is easy to build but requires excess capacity on operational database servers.

From data warehouse to data mining

- Data warehousing is the process of constructing and using data warehouses.
- The construction of a data warehouse requires data cleaning, data integration, and data consolidation.
- Many organizations use this information to support business decision-making activities
 - Including Increasing customer focus, which includes the analysis of customer buying patterns (such as buying preference, buying time, budget cycles, and appetites for spending)
 - repositioning products and managing product portfolios by comparing the performance of sales by quarter, by year, and by geographic regions in order to fine-tune production strategies;
 - analyzing operations and looking for sources of profit; and
 - Managing customer relationships, making environmental corrections, and managing the cost of corporate assets

End of Chapter
?

CHAPTER Two

DATA PREPROCESSING

What is Data Pre-processing?

- Data Preprocessing is a technique that is used to convert the raw data into a clean dataset.
- It is used to transform the raw data in a useful and efficient format.
- Data preprocessing is used for representing complex structures with attributes, **discretization of continuous attributes**, which is converting continuous attributes to categorical, and **dealing with missing** and **unknown attribute values**.
- Various visualization techniques provide valuable help in data preprocessing.

Why process the data?

- ❖ Data in the real world may be,
 - ✓ *Inaccurate data (missing data)*
 - lacking *attribute values*, lacking certain *attributes of interest*, or containing only aggregate data.
 - ✓ *The presence of noisy data (erroneous data and outliers(exceptions))*
 - ✓ *Inconsistent*: containing mismatches in codes or names when data is integrated from multiple source.
- ❖ No quality data, no quality mining results!
- ❖ In other words, whenever the data is gathered from different sources it is collected in raw format which is not feasible for the analysis.
- ***Quality decisions must be based on quality data.***

Why can Data be Incomplete?

- ❖ Attributes of interest are not **available** (e.g., customer information for sales transaction data)
- ❖ Data were not considered important at the **time** of transactions, so they were not recorded!
- ❖ Data not recorder because of **misunderstanding** or malfunctions
- ❖ Data may have been recorded and **later** deleted!
- ❖ **Missing/unknown** values for some data.

Why can Data be Noisy/Inconsistent?

- ✓ Faulty instruments for data collection
- ✓ Human or computer errors
- ✓ Errors in data transmission
- ✓ Technology limitations (e.g., sensor data come at a faster rate than they can be processed).
- ✓ Inconsistencies in naming conventions or data codes (e.g., 2/5/2002 could be 2 May 2002 or 5 Feb 2002).
- ✓ Duplicate tuples, which were received twice should also be removed.

Major Tasks in Data Preprocessing

❖ *Data cleaning*

- Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies

❖ *Data integration*

- Integration of multiple databases, data cubes, or files

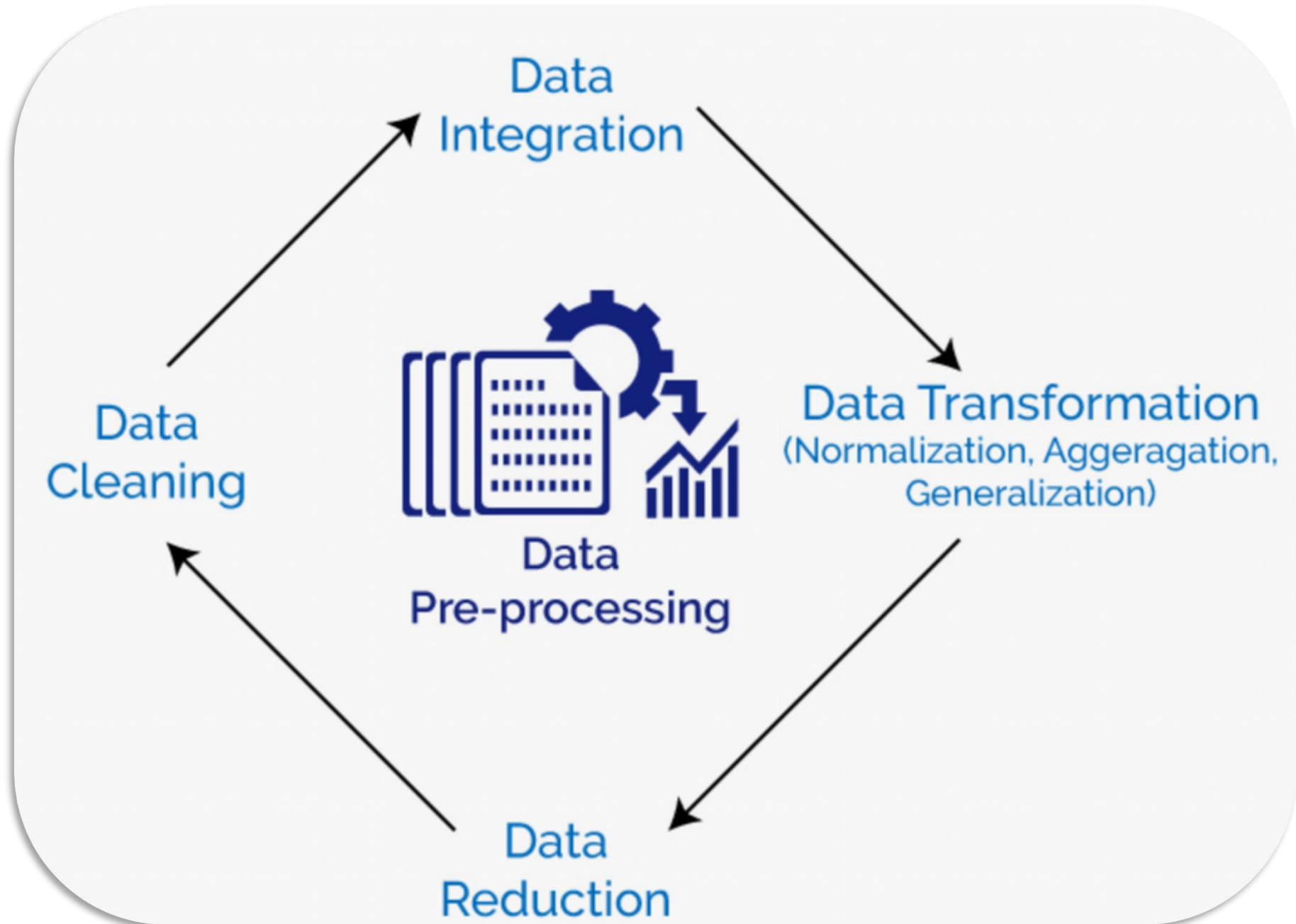
❖ *Data transformation*

- Normalization , aggregation and Generalization

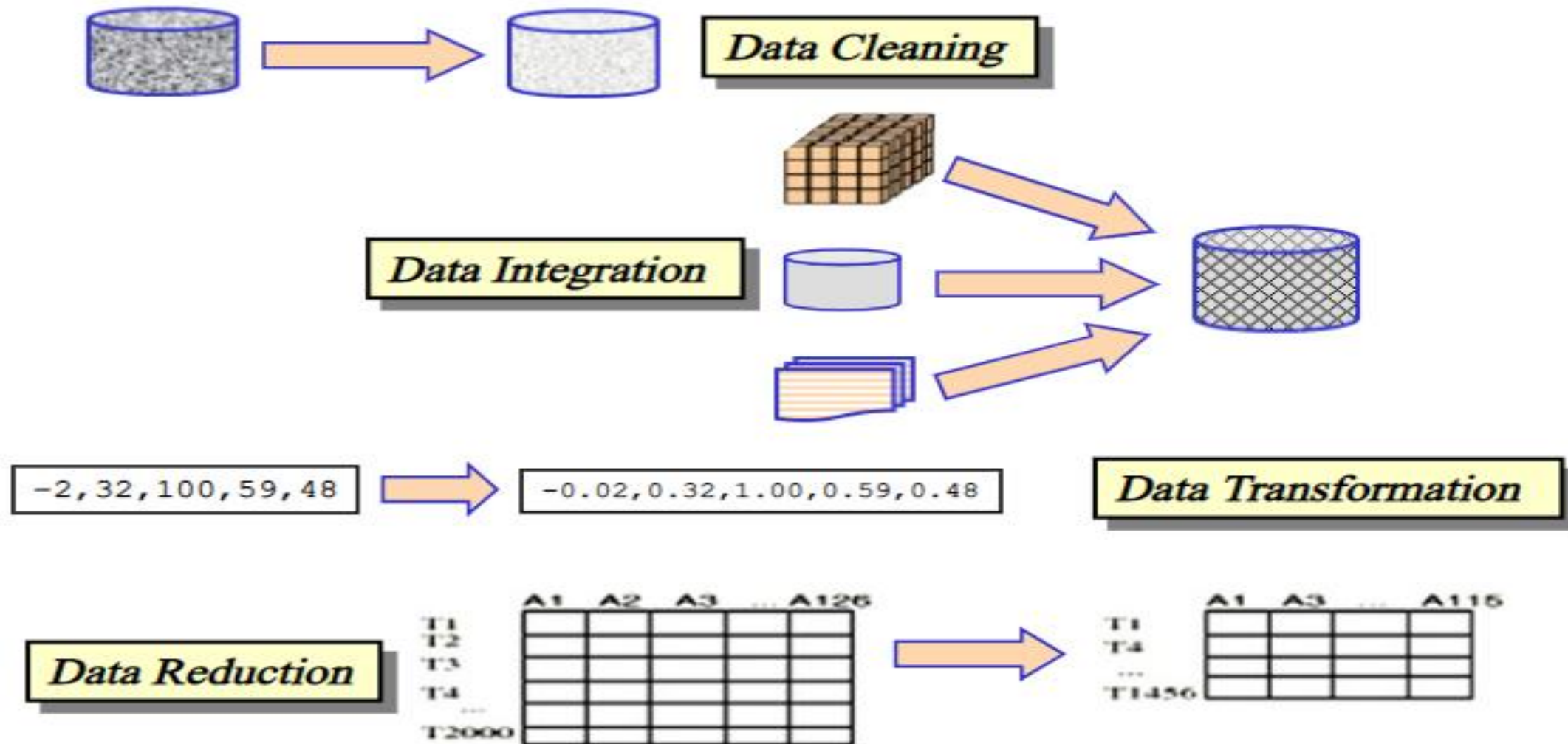
❖ *Data reduction*

- Obtains reduced representation in volume but produces the same or similar analytical results
- Feature selection (i.e., relevant subset of all attributes which are important to our analysis are selected).

❖ *Data discretization*:-Part of data reduction but with particular importance, especially for numerical data



Forms of data preprocessing



Data cleaning

- ❖ Real-world data tend to be **incomplete, noisy, and inconsistent**.
- ❖ Data cleaning (or data cleansing) routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data.

❖ **Missing Values:** How can you go about filling in the missing values for this attribute?

1. **Ignore the tuple:** This is usually done when the class label is missing (assuming the mining task involves classification). This method is not very effective, unless the tuple contains several attributes with missing values.

2. **Fill in the missing value manually:** In general, this approach is time-consuming and may not be feasible given a large data set with many missing values.

3. **Use a global constant to fill in the missing value:** Replace all missing attribute values by the same constant, such as a label like “Unknown” or $-\infty$.

5. **Use the attribute mean for all samples belonging to the same class as the given tuple**

For example, if classifying customers according to credit risk, replace the missing value with the average income value for customers in the same credit risk category as that of the given tuple.

6. **Use the most probable value to fill in the missing value**

Noisy Data

“What is noise?”

- Noise is a random error or variance in a measured variable.
- Given a numerical attribute such as, price, how can we “smooth” out the data to remove the noise?
- Let’s look at the following data smoothing techniques:
- the following data smoothing techniques:

Binning:-

- ✓ Binning methods smooth a sorted data value by consulting its “neighborhood,” that is, the values around it.
- ✓ The sorted values are dis-tributed into a number of “buckets,” or bins.
- ✓ Because binning methods consult the neighborhood of values, they perform local smoothing

cont.....

A given data e.g price are first sorted and then **partitioned into equal-frequency bins** of size 3 (i.e., each bin contains three values).

Then it will smooth using the following methods

Sorted data for price (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24

Bin 3: 25, 28, 34

Smoothing by bin means:

Bin 1: 9, 9, 9

Bin 2: 22, 22, 22

Bin 3: 29, 29, 29

➤ **Smoothing by bin boundaries:**

➤ Bin 1: 4, 4, 15

➤ Bin 2: 21, 21, 24

➤ Bin 3: 25, 25, 34

➤ Smoothing by bin Median??

■ ■ ■ ■

- **Regression**:-Data smoothing can also be done by conforming data values to a function, a technique known as regression.
- Linear regression involves finding the “best” line to fit two attributes (or variables), so that one attribute can be used to predict the other.
- Multiple linear regression is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensional surface.

Outlier analysis

❖ An outlier is a data value that is very different from the rest of the data.

Example: Data = 5, 7, 8, 9, 10, 100

→ 100 is an outlier (too far from others)

Outlier analysis is the process of identifying and handling data values that are significantly different from the rest of the dataset.

Data integration

- ❖ **Data integration** combines data from multiple sources (database, data warehouse, files & sometimes from non-electronic sources) into a coherent store.
- ❖ Some of the **issues** are:
 - ✓ Different formats and structures
 - ✓ Conflicting and redundant data
 - ✓ Data at different levels

Data Integration: Formats

- ❖ Not everyone uses the same format.
 - Schema integration: e.g., A.cust-id $\equiv$ B.cust-#
 - ✓ Integrate metadata from different sources
- ❖ Dates are especially problematic:
 - ✓ 12/19/97
 - ✓ 19/12/97
 - ✓ 19/12/1997
 - ✓ 19-12-97
 - ✓ Dec 19, 1997
 - ✓ 19 December 1997
 - ✓ 19th Dec. 1997
- ❖ Are you frequently writing money as:
 - Birr 200, Br. 200, 200 Birr, ...

Data Integration: Inconsistent

- Inconsistent data:-containing discrepancies in codes or names, which is also the problem of lack of standardization / naming conventions.
- e.g.
 - ✓ *Age*="26" vs. *Birthday*="03/07/1986"
 - ✓ Some use "1,2,3" for rating; others "A, B, C"
- Discrepancy between duplicate records : inconsistencies or differences found in duplicate or identical records within a dataset

ID	Name	City	State
1	Ministry of Transportation	Addis Ababa	Addis Ababa region
2	Ministry of Finance	Addis Ababa	Addis Ababa administration
3	Office of Foreign Affairs	Addis Ababa	Addis Ababa regional administration

Data Integration: Conflicting Data

- ❖ Detecting and resolving data value conflicts
 - For the same real world entity, attribute values from different sources are different
 - Possible reasons: different representations, different scales,
e.g., American vs. British units
 - ✓ weight measurement: KG or pound
 - ✓ Height measurement: meter or inch
- ❖ Information source #1 says that Alex lives in Bahirdar
 - Information source #2 says that Alex lives in Mekele
- ❖ What to do?
 - ✓ Use both (He lives in both places)
 - ✓ Use the most recently updated piece of information
 - ✓ Use the “most trusted” information
 - ✓ Flag row to be investigated further by hand
 - ✓ Use neither (We’d rather be incomplete than wrong)

Handling Redundancy in Data Integration

- Redundant data occur often when integration of multiple databases
 - ❖ **Object identification:** -The same attribute or object may have different names in different databases
 - ❖ **Derivable data:**-One attribute may be a “derived” attribute in another table, e.g., annual revenue, age
- Redundant attributes may be able to be detected by *correlation analysis and covariance analysis*
- **Careful integration** of the data from multiple sources may help reduce/avoid redundancies, inconsistencies and improve mining speed and quality

Data Transformation

- ❖ In data transformation, the data are transformed or consolidated into forms appropriate for mining.
- ❖ Data transformation can involve the following:
 - **Smoothing**, which works to remove noise from the data.
 - ✓ Such techniques include **binning, regression, and clustering.**
 - **Aggregation**, where summary or aggregation operations are applied to the data.
 - ✓ For example, the daily sales data may be aggregated so as to compute monthly and annual total amounts.
 - ✓ This step is typically used in constructing a data cube for analysis of the data at multiple granularities.

Data Transformation

- **Generalization of the data**, where low-level or “primitive” (raw) data are replaced by higher-level concepts through the use of concept hierarchies.
- ✓ For example, categorical attributes, like street, can be generalized to higher-level concepts, like city or country. Similarly, values for numerical attributes, like age, may be mapped to higher-level concepts, like youth, middle-aged, and senior.

Cont..

- **Normalization**, where the attribute data are scaled so as to fall within a small specified range, such as 0.0 to 1.0. is particularly useful for classification algorithms involving neural networks, or distance measurements such as nearest-neighbor classification and clustering.
- Additional methods also:
 - ✓ min-max normalization
 - ✓ z-score normalization
- **Attribute construction (or feature construction)**, where new attributes are constructed and added from the given set of attributes to help the mining process.
- For example, we may wish to add the attribute area based on the attributes height and width.

Data reduction

❖ **Techniques** can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data.

❖ **Strategies for data reduction include the following:**

1. **Data cube aggregation**, where aggregation operations are applied to the data in the construction of a data cube. It involves aggregation operations like sum, count, average to the data during the construction of a data cube.

2. **Attribute subset selection**, where irrelevant, weakly relevant, or redundant attributes or dimensions may be detected and removed.

3. **Dimensionality reduction**, where encoding mechanisms are used to reduce the data set size.

Cont...

4. Numerosity reduction, where the data are replaced or estimated by alternative, smaller data representations such as parametric models (which need store only the model parameters instead of the actual data)

e.g mean or nonparametric methods such as clustering, sampling, and the use of histograms.

5. Discretization and concept hierarchy generation,

where raw data values for attributes are replaced by ranges or higher conceptual levels.

Discretization

- ❖ **Discretization:** Reduce data size by dividing the range of a continuous attribute into intervals.
- ❖ Interval labels can then be used to replace **actual data values**
 - Discretization can be performed recursively on an attribute using method such as
 - ✓ **Binning:** divide values into intervals
 - ✓ **Concept hierarchy climbing:** organizes concepts (i.e., attribute values) hierarchically

Two types of attributes:

❖ **Categorical** :-alphabet

- ✓ **Nominal**:-values from an unordered set: such as colors, name , sex
- ✓ **Ordinal**:-values from an ordered set, such as Rankings, Grades, Height

❖ **Continuous** :- real numbers

Binning

Equal Width Binning

- ✓ Equal Width Binning is a data mining technique that divides a continuous variable into bins of equal widths or intervals.
- ✓ This approach aims to group data points into bins with the same range or width.

Example

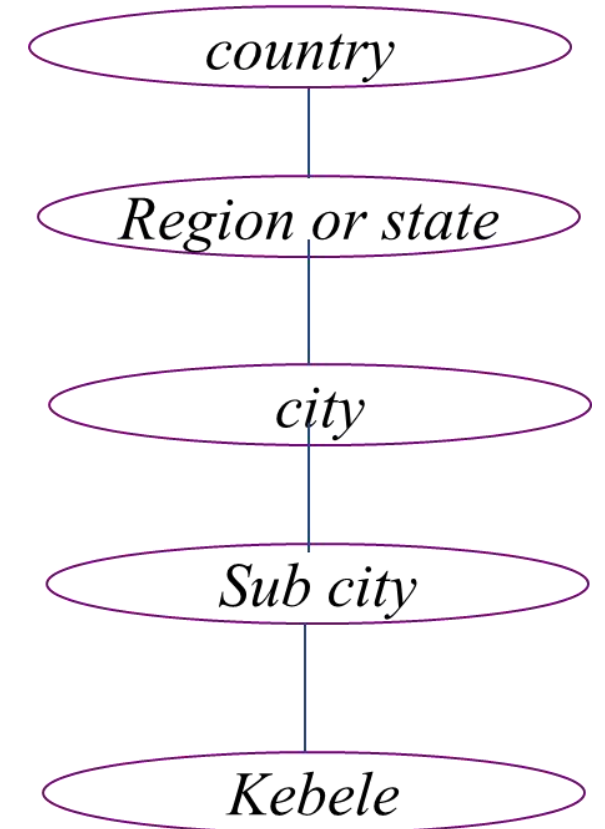
- Dataset points: 10, 15, 18, 20, 31, 34, 41, 46, 51, 53, 54
- ❖ Step 1: Sort the dataset in ascending order: 10, 15, 18, 20, 31, 34, 41, 46, 51, 53, 54.
- ❖ Step 2: Determine the width of each bin using the formula: $w = (\text{max} - \text{min}) / N$,
where w is the width, **max is the maximum value**, **min is the minimum value**, and **N is the number of bins**.
- ❖ Eg. the minimum value is **10**, the maximum value is **54**,
- ❖ We choose four bins. So, the width (**w**) is calculated as **$(54 - 10) / 4 = 11$** . Now we have to add value to each bin which would be as follows:
 - BIN 1 : [10, 15, 18, 20]
 - BIN 2 : [31]
 - BIN 3 : [34, 41]
 - BIN 4 : [46, 51, 53, 54]
- ✓ BIN 1 : [lower bound , upper bound] = [(min) , (min + w -1)] = [10, 20]
- ✓ BIN 2 : [lower bound , upper bound] = [(min + w) , (min + 2w -1)] = [21, 31]
- ✓ BIN 3 : [lower bound , upper bound] = [(min + 2w) , (min + 3w -1)] = [32, 42]
- ✓ BIN 4 : [lower bound , upper bound] = [(min + 3w) , (max)] = [43, 54]

Equal Frequency Binning

- This method major goal is to ensure every bin has a comparable amount of data points.
- This helps us to present the information in a balanced manner.
- **Dataset:- 10,15, 18, 20, 31, 34, 41, 46, 51, 53, 54, 60**
- If we have unsorted data
 - ✓ Convert it into sorted data
 - ✓ Find the frequency.
 - ✓ To calculate the frequency, we can use the formula **a total number of data points/number of bins.**
- ❖ In this case, the total number of data points is 12, & the number of bins required is 3.
- ❖ **Therefore, the frequency comes out to be 4.**
- ❖ **Now let's add values into the bins.**
 - ✓ BIN 1: 10, 15, 18, 20
 - ✓ BIN 2: 31, 34, 41, 46
 - ✓ BIN 3: 51, 53, 54, 60

Concept Hierarchy Generalization

- ❖ organizes concepts (i.e., attribute values) hierarchically and is usually associated with each dimension in a data warehouse
 - Concept hierarchy formation: **Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for *age*) by higher level concepts (such as *youth*, *adult*, or *senior*)**
- ❖ Concept hierarchies can be explicitly specified by domain experts and/or data warehouse designers Concept hierarchy can be automatically formed by the analysis of the number of distinct values.
- ❖ E.g., for a set of attributes: {Kebele, city, state, country}
- ❖ **For numeric data, use discretization methods.**



Thank You!!